

What is “aphasia-friendly” about aphasia friendly materials?

A sandbox series presentation

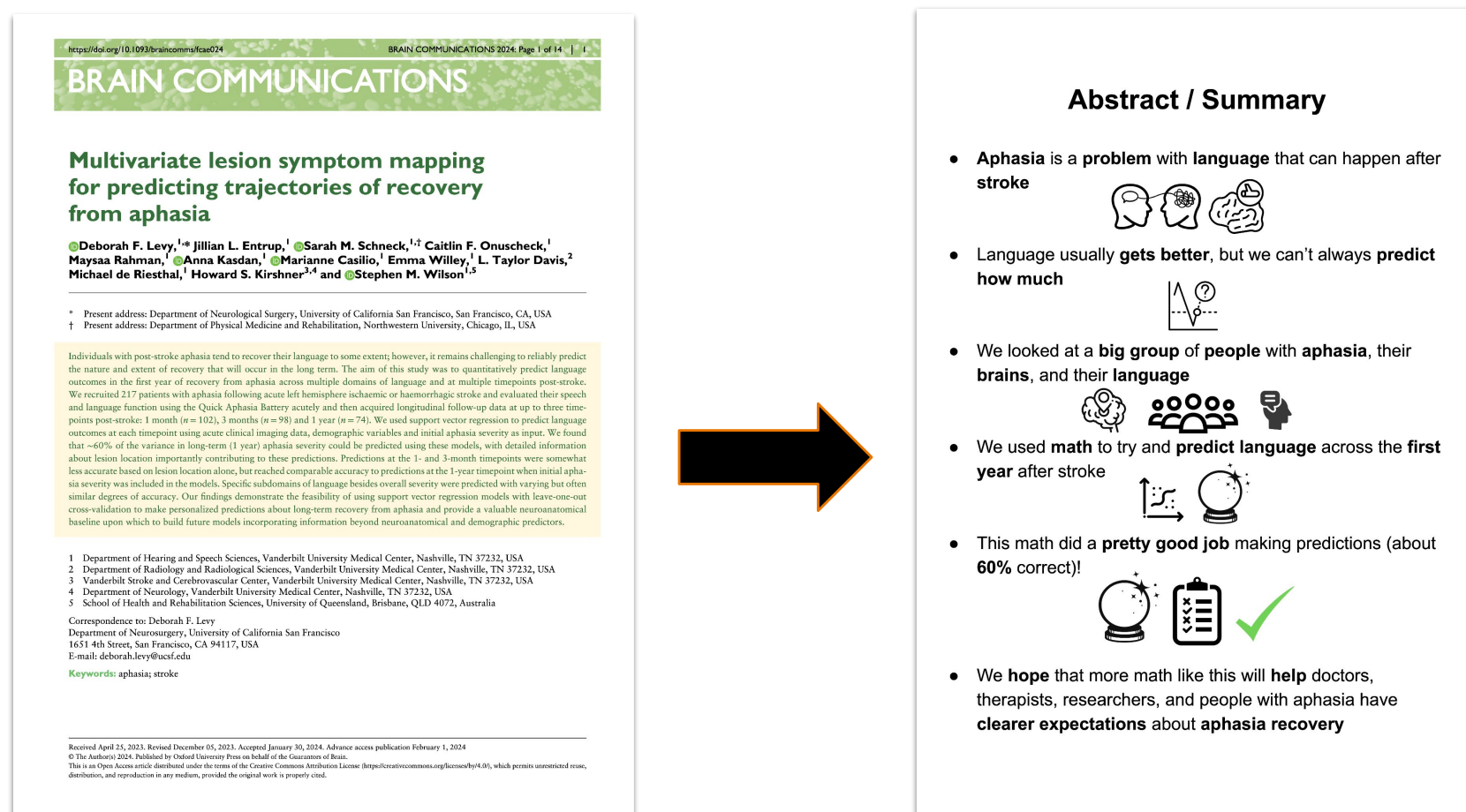
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The problem

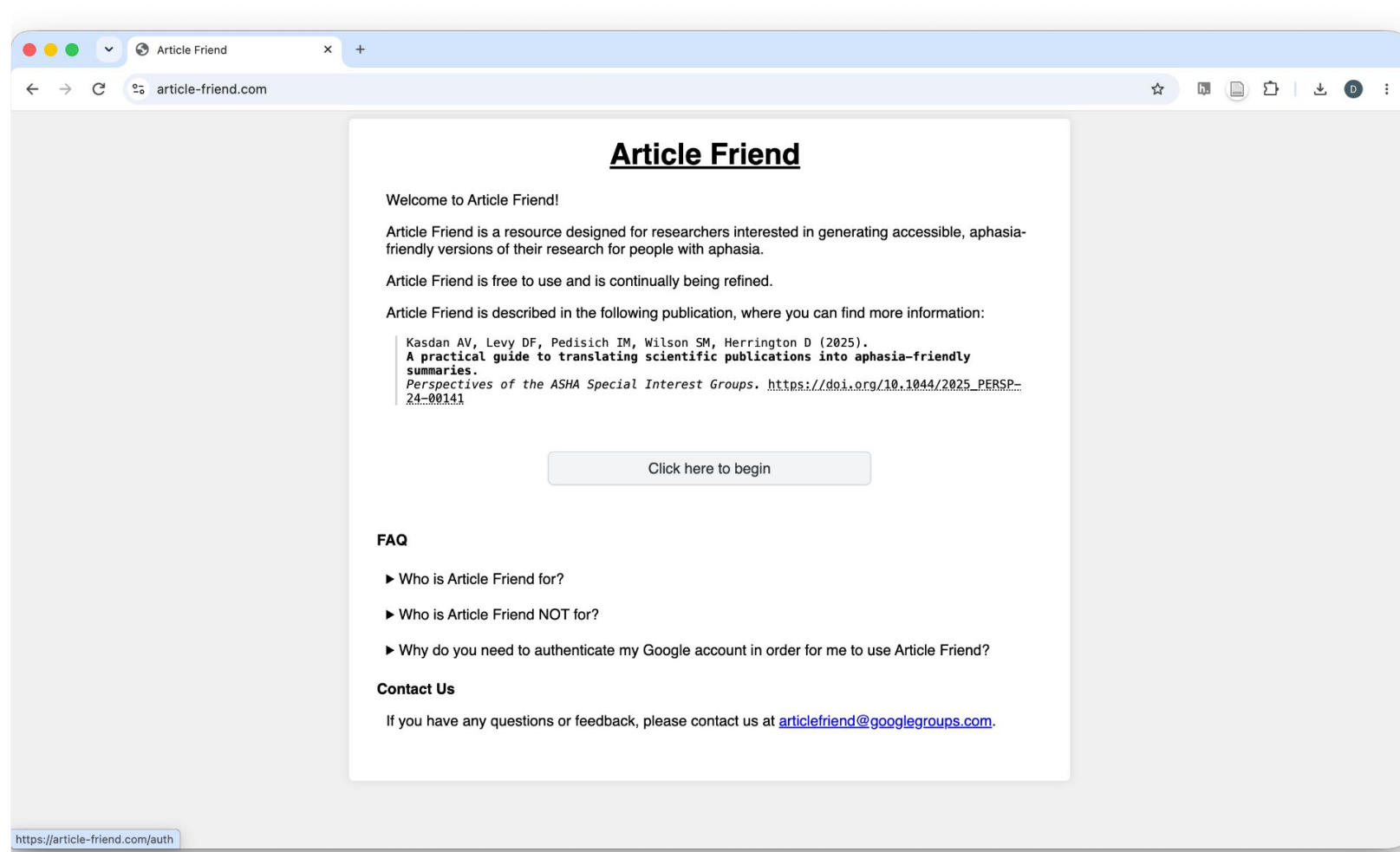
- Research about aphasia is largely not accessible to people with aphasia
- Some recent scientific papers have published aphasia-friendly aids along with the traditional research articles [e.g. 1, 2]
- However,
 - creating these aids can take time, and
 - knowing how to make these aids most useful to people with aphasia is difficult without their input

The project

- We recently published an initial how-to guide for generating aphasia-friendly aids to research articles [3]



- This paper introduced *Article Friend*, a tool that uses large language models (LLMs) to jump-start the process of aid creation for researchers



- We want to test, empirically, whether these tools “work” for researchers and people with aphasia, thereby expanding the evidence base for this practice
- Here, we share our plans for empirically evaluating these research aids and tools, proceeding in two parts:
 - Part 1 (in progress): LLM evaluation
 - Part 2 (planned): Stakeholder surveys

The progress (LLMs)

Automatically evaluate the quality of LLM-generated Article Friend output

Materials

- Abstracts from most cited papers from *Neurobiology of Language* (n=7) and *Journal of Speech, Language, and Hearing Research* (n=7) in the last 5 years

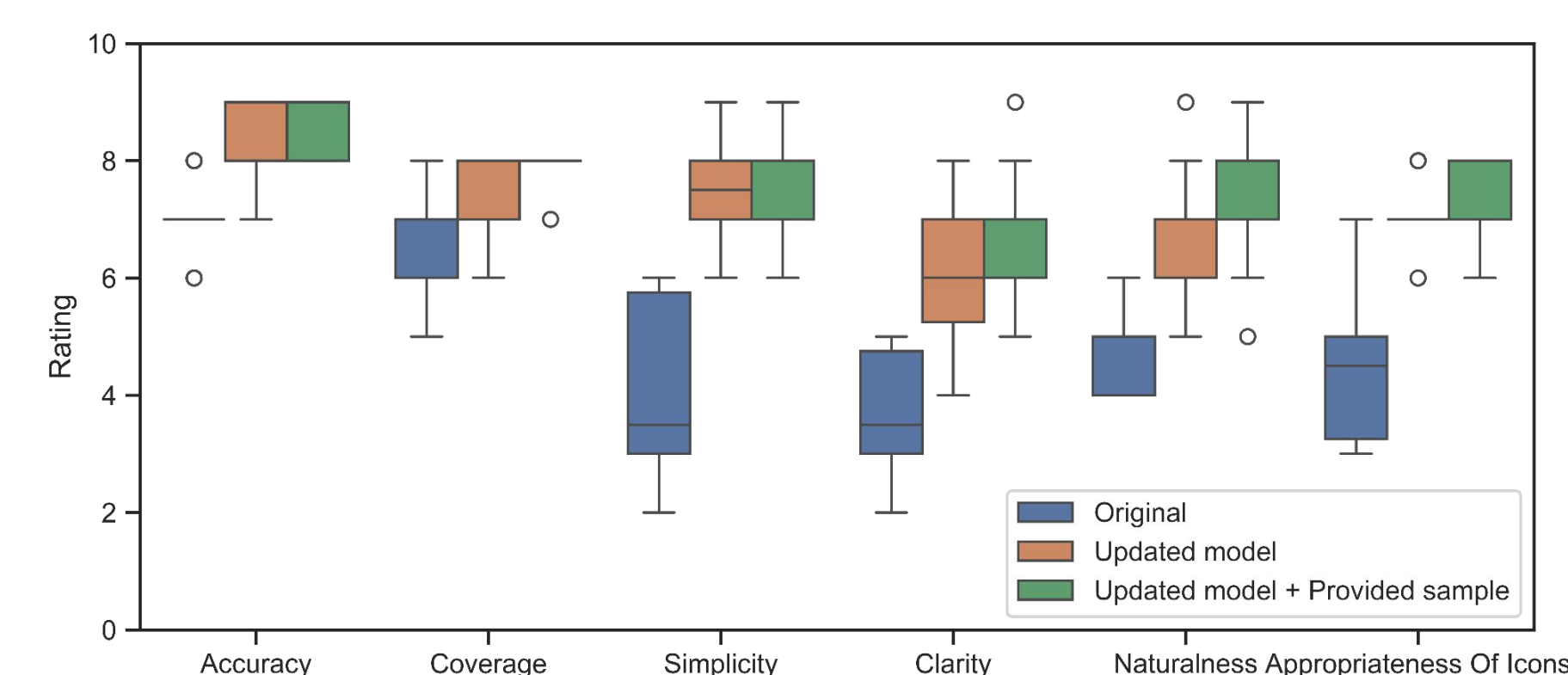
Methods

- Generate simplified abstracts using *Article Friend* (LLM provider: OpenAI)
- Compare original vs. updated (e.g. newer models, altered prompts) versions of *Article Friend*
- Separate agent (LLM provider: Anthropic) rates each abstract on 6 dimensions
 - Accuracy
 - Coverage
 - Simplicity
 - Clarity
 - Naturalness
 - Appropriateness of icons

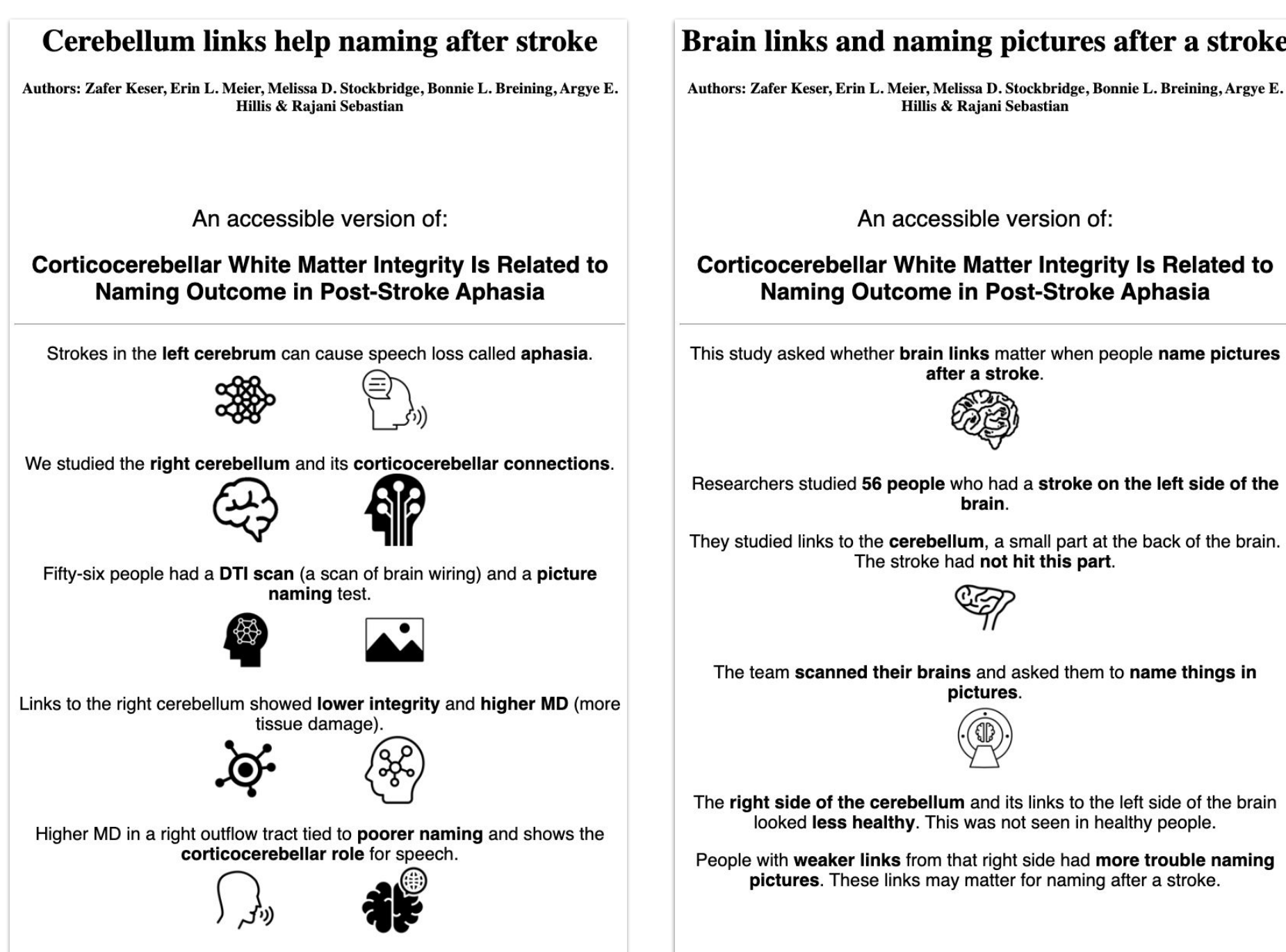
Original vs. updated version of *Article Friend*

- Updated model, added example to prompt, used stricter prompting language

Results



- Largest improvements resulted from model update, but careful editing of the prompt and providing relevant examples also had noticeable effects
- Creating an automated evaluation loop will bring more stability to the previously tenuous / subjective process of updating the tool



- The next step for *Article Friend* will be to add user interactivity, allowing for choices between phrases and manual selection from an array of icons

The plan (Stakeholders)

Present individuals with aphasia with research abstracts that vary along parameters of aphasia-friendly features

Materials

- Same 14 abstracts from the LLM portion of the project
- Abstract versions will include the following:
 - Original published version
 - Simplified language
 - Simplified language + keyword bolding
 - Simplified language + icons
 - Simplified language + photographs
 - Simplified language + keyword bolding + icons
 - Simplified language + keyword bolding + photographs

Participants

- 40 individuals with aphasia due to stroke or brain injury

Planned outcomes and analyses

- Outcomes
 - Reading comprehension accuracy
 - Participant preference for abstract format
- Analyses: Linear mixed effects models that account for aphasia severity (participants' most recent WAB-AQ or QAB score)

Discussion

- Through these efforts, we hope to learn:
 - How people with aphasia process different forms of information
 - How to best aid comprehension and accessibility of research
 - How LLMs can support researchers in creating materials that enhance accessibility of their publications
- Previous work has addressed questions about text simplification, iconographic vs. photographic visual aids, and line spacing / font style for aiding comprehension and improving reader experience for people with aphasia [e.g. 4, 5, 6].
- We hope that this work will increase both the evidence base for and the ease of creating aphasia-friendly aids to research articles.

References

- [1] Levy et al., 2022. *Perspectives of the ASHA Special Interest Groups*. [2] Casilio et al., 2025. *Brain*. [3] Kasdan et al., 2025. *Perspectives of the ASHA Special Interest Groups*. [4] van der Steldt et al., 2026. *BMJ Open*. [5] Rose et al., 2012. *Int J Speech-Lang Pathol* 2012. [6] Wilson & Read, 2016. *Int J Lang Commun Disord*.

Acknowledgments

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